

SKILL-5: SHELL QUOTING MECHANISMS (SINGLE & DOUBLE QUOTES)

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Task 1 — Single Quotes in Shell

Objective: To apply single quotes in Bash and understand that everything inside ' ' is treated literally, variable expansion does not happen, spaces are preserved, special characters are treated as normal characters, the quoted string can be stored in a variable, and parsing can be validated using echo, printf, and set -x.

Key Steps & Execution Breakdown:

- **Step 1-3 (Variables & Single Quotes):** Created variable name="Aruntej". Running echo \$name outputs Aruntej, whereas echo '\$name' outputs literally \$name because single quotes prevent variable expansion.
- **Step 4-5 (Literal Content & Spaces):** Expressions like echo 'Hello \$name' and echo 'Hello * ? [] { }' preserve special characters as normal text. Spaces inside single quotes are preserved without word splitting.
- **Step 6-8 (Assignment & Validation):** Stored quoted strings in variables (e.g., message='Welcome to Ubuntu Linux'). Used Bash tracing with set -x to inspect how Bash executes quoted commands.
- **Step 9 (Edge Cases):** Tested empty quotes ("), numbers/symbols, command substitution ('Today is \$(date)' which remains literal), and handling nested quotes.

Execution Screenshots (Task 1):

```

aruntej@LAPTOP-DGRT8C5G:~$ pwd
/home/aruntej
aruntej@LAPTOP-DGRT8C5G:~$ name="Aruntej"
aruntej@LAPTOP-DGRT8C5G:~$ echo $name
Aruntej
aruntej@LAPTOP-DGRT8C5G:~$ echo '$name'
$name
aruntej@LAPTOP-DGRT8C5G:~$ echo 'Hello $name'
Hello $name
aruntej@LAPTOP-DGRT8C5G:~$ echo 'Hello * 7 [ ] {}'
Hello * 7 [ ] {}
aruntej@LAPTOP-DGRT8C5G:~$ echo 'Hello      Ubuntu      World'
Hello      Ubuntu      World
aruntej@LAPTOP-DGRT8C5G:~$ message='Welcome to Ubuntu Linux'
aruntej@LAPTOP-DGRT8C5G:~$ echo "$message"
Welcome to Ubuntu Linux
aruntej@LAPTOP-DGRT8C5G:~$ message='$name is learning Ubuntu'
aruntej@LAPTOP-DGRT8C5G:~$ echo "$message"
$name is learning Ubuntu
aruntej@LAPTOP-DGRT8C5G:~$ set -x
echo '$name'
set -x
++ __systemd_osc_context_psh
++ '[' -n 1c24e9b8-8463-478b-99c9-22afa8f4381f ']'
++ __systemd_osc_context_common
+++ '[' -f /etc/machine-id ']'
+++ printf '%s\n' 'd003082ce67e43f8f7017759e4d73db3'
+++ printf '%s\n' 'user=255;hostname=255;bootid=363;pid=20d' aruntej LAPTOP-DGRT8C5G 3387f977-039f-4b1b-8c4c-ecce8ea3ff2f
+++ echo /home/aruntej
+++ sed -e 's/\\/\x5c/g' -e 's/;/\x3b/g' -e 's/[[:cntrl:]]/\w/g'
++ printf '\033[3080;start=645s;type=command;cwd=255\033\ 1c24e9b8-8463-478b-99c9-22afa8f4381f ' 'machineid=d003082ce67e43f8f7017759e4d73db3;user=aruntej;hostname=LAPTOP-DGRT8C5G;bootid=3387f977-039f-4b1b-8c4c-ecce8ea3ff2f;pid=0000000000000000775' /home/aruntej
++ echo '$name'
$name
++ __systemd_osc_context_psh
++ '[' -n 1c24e9b8-8463-478b-99c9-22afa8f4381f ']'
++ __systemd_osc_context_common
+++ '[' -f /etc/machine-id ']'
+++ printf '%s\n' 'd003082ce67e43f8f7017759e4d73db3'
+++ printf '%s\n' 'user=255;hostname=255;bootid=363;pid=20d' aruntej LAPTOP-DGRT8C5G 3387f977-039f-4b1b-8c4c-ecce8ea3ff2f
+++ __systemd_osc_context_escape /home/aruntej
+++ echo /home/aruntej
+++ sed -e 's/\\/\x5c/g' -e 's/;/\x3b/g' -e 's/[[:cntrl:]]/\w/g'
++ printf '\033[3080;start=645s;type=command;cwd=255\033\ 1c24e9b8-8463-478b-99c9-22afa8f4381f ' 'machineid=d003082ce67e43f8f7017759e4d73db3;user=aruntej;hostname=LAPTOP-DGRT8C5G;bootid=3387f977-039f-4b1b-8c4c-ecce8ea3ff2f;pid=0000000000000000775' /home/aruntej
+ set -x

```

```

aruntej@LAPTOP-DGRT8C5G:~$ echo ''
aruntej@LAPTOP-DGRT8C5G:~$ echo '12345 ! @ # $ % ^ & *'
12345 ! @ # $ % ^ & *
aruntej@LAPTOP-DGRT8C5G:~$ echo 'Today is $(date)'
Today is $(date)
aruntej@LAPTOP-DGRT8C5G:~$ echo 'Aruntej's Ubuntu'
> ^C
aruntej@LAPTOP-DGRT8C5G:~$ echo "Aruntej's Ubuntu"
Aruntej's Ubuntu
aruntej@LAPTOP-DGRT8C5G:~$ name="Aruntej"

echo '$name'
echo 'Hello Ubuntu'
echo 'Hello      Linux'
echo 'Today is $(date)'

message='This is a quoted string'
echo "$message"
$name
Hello Ubuntu
Hello      Linux
Today is $(date)
This is a quoted string
aruntej@LAPTOP-DGRT8C5G:~$ |

```

```

aruntej@LAPTOP-DGRT8C5G:~$ name="Aruntej"
course="Computer Science"
aruntej@LAPTOP-DGRT8C5G:~$ echo "$name"
Aruntej
Computer Science
aruntej@LAPTOP-DGRT8C5G:~$ echo "My name is $name"
My name is Aruntej
aruntej@LAPTOP-DGRT8C5G:~$ echo "I am studying $course"
I am studying Computer Science
aruntej@LAPTOP-DGRT8C5G:~$ message="Hello      Ubuntu      Linux"
aruntej@LAPTOP-DGRT8C5G:~$ echo "$message"
Hello      Ubuntu      Linux
aruntej@LAPTOP-DGRT8C5G:~$ message="Hello Ubuntu Linux"
aruntej@LAPTOP-DGRT8C5G:~$ printf '<%s>\n' $message
<Hello>
<Ubuntu>
<Linux>
aruntej@LAPTOP-DGRT8C5G:~$ printf '<%s>\n' "$message"
<Hello Ubuntu Linux>
aruntej@LAPTOP-DGRT8C5G:~$ echo "Current date: $(date)"
Current date: Tue Sep 22 04:09:05 UTC 2026
aruntej@LAPTOP-DGRT8C5G:~$ user="Aruntej"
place="Hyderabad"
aruntej@LAPTOP-DGRT8C5G:~$ echo "User: $user, Location: $place"
User: Aruntej, Location: Hyderabad

```

Task 2 — Double Quotes in Shell

Objective: To apply double quotes and understand that spaces are preserved, variables are expanded, command substitution works, nested tokens can be handled, quoted commands can be tested, and parsing can be validated.

Key Steps & Execution Breakdown:

- **Step 1-4 (Variable Expansion & Spaces):** Defined name="Aruntej" and course="Computer Science". Double quotes allow variable expansion (e.g., echo "My name is \$name") while preserving spaces and preventing unwanted word splitting compared to unquoted variables.
- **Step 5-8 (Command Substitution & Nested Tokens):** Executed command substitution like echo "Current date: \$(date)" and echo "Files in current directory: \$(ls)". Handled directories with spaces such as mkdir "Ubuntu Lab" and navigating with cd.
- **Step 9-14 (Validation & Edge Cases):** Used printf '<%s>\n' "\$text" to prove quoted variables are treated as a single argument, tested empty variables (empty=""), and validated parsing using Bash tracing (set -x).

Execution Screenshots (Task 2):

```
aruntej@LAPTOP-DGRT8C5G:~$ echo "Files in current directory: $(ls)"
Files in current directory: Hello.c
in current directory: Hello.c

aruntej@LAPTOP-DGRT8C5G:~$ mkdir "Ubuntu Lab"
aruntej@LAPTOP-DGRT8C5G:~$ ls
Hello.c          experiment-3.c    program4.c       signal1.c
'Ubuntu Lab'     experiment-4.c    program5.c       skill-1.c
a.out            experiment-4.c    program5.c       skill-1.c
client           fork.c           program6.c       skill-2.c
client.c         fork.c           program6.c       skill-2.c
client1.c        input.txt        program7.c       source.txt
destination.txt  notes.txt        program7.c       systemcall
exp1.c           ossp             program8.c       systemcall.c
exp2             output.txt       program8.c       systemcall2
exp2.c           producerconsumer program9.c       systemcall2.c
exp3             producerconsumer program9.c       upipe.fifo
exp3.c           program1.c       readfiles        upipe.txt
experiment       program1.c       readfiles.c      zombie
experiment-1     program2.c       rreadfiles.c    zombie.c
experiment-1.c   program2.c       sample.txt      zombie_fixed
experiment-2     program3.c       serve            zombie_fixed.c
experiment-2.c   program3.c       serve.c
experiment-3     program4         signal
experiment-3.c   program4         signal.c

aruntej@LAPTOP-DGRT8C5G:~$ cd "Ubuntu Lab"
aruntej@LAPTOP-DGRT8C5G:~/Ubuntu Lab$ ^C
aruntej@LAPTOP-DGRT8C5G:~/Ubuntu Lab$ cd ..
aruntej@LAPTOP-DGRT8C5G:~$ text="Ubuntu Linux Lab"

printf '<%s>\n' "$text"
<Ubuntu Linux Lab>
aruntej@LAPTOP-DGRT8C5G:~$ printf '<%s>\n' $text
<Ubuntu>
<Linux>
<Lab>
```

```

aruntej@LAPTOP-DGRT8C5G:~$ set -x
name="Aruntej"
echo "Hello $name"
set +x
++ _systemd_osc_context_ps0
++ '[' -n 8a10e1e3-88d3-4e27-b619-74f1936b4372 ']'
+++ _systemd_osc_context_common
+++ '[' -f /etc/machine-id ']'
+++ printf 'machineid=%,36s' d003082ce67e438f87017759e4d73db3
+++ printf 'user=%,255s;hostname=%,255s;bootid=%,36s;pid=%,20d' aruntej LAPTOP-DGRT8C5G 3387f977-039f-4b1b-8c4c-ecee8ea3ff2f 1048
+++ _systemd_osc_context_escape /home/aruntej
+++ echo /home/aruntej
+++ sed -e 's/\\/\x5c/g' -e 's/;/\x3b/g' -e 's/[[:cntrl:]]/\0/g'
+++ printf '\033]3000;start=%,64s%;type=command;cwd=%,255s\033\'' 8a10e1e3-88d3-4e27-b619-74f1936b4372 'machineid=d003082ce67e438f87017759e4d73db3;user=aruntej;hostname=LAPTOP-DGRT8C5G;bootid=3387f977-039f-4b1b-8c4c-ecee8ea3ff2f;pid=00000000000000001048' /home/aruntej
+ name=Aruntej
++ _systemd_osc_context_ps0
++ '[' -n 8a10e1e3-88d3-4e27-b619-74f1936b4372 ']'
+++ _systemd_osc_context_common
+++ '[' -f /etc/machine-id ']'
+++ printf 'machineid=%,36s' d003082ce67e438f87017759e4d73db3
+++ printf 'user=%,255s;hostname=%,255s;bootid=%,36s;pid=%,20d' aruntej LAPTOP-DGRT8C5G 3387f977-039f-4b1b-8c4c-ecee8ea3ff2f 1048
+++ _systemd_osc_context_escape /home/aruntej
+++ echo /home/aruntej
+++ sed -e 's/\\/\x5c/g' -e 's/;/\x3b/g' -e 's/[[:cntrl:]]/\0/g'
+++ printf '\033]3000;start=%,64s%;type=command;cwd=%,255s\033\'' 8a10e1e3-88d3-4e27-b619-74f1936b4372 'machineid=d003082ce67e438f87017759e4d73db3;user=aruntej;hostname=LAPTOP-DGRT8C5G;bootid=3387f977-039f-4b1b-8c4c-ecee8ea3ff2f;pid=00000000000000001048' /home/aruntej
+ echo 'Hello Aruntej'
+ Hello Aruntej
++ _systemd_osc_context_ps0
++ '[' -n 8a10e1e3-88d3-4e27-b619-74f1936b4372 ']'
+++ _systemd_osc_context_common
+++ '[' -f /etc/machine-id ']'
+++ printf 'machineid=%,36s' d003082ce67e438f87017759e4d73db3
+++ printf 'user=%,255s;hostname=%,255s;bootid=%,36s;pid=%,20d' aruntej LAPTOP-DGRT8C5G 3387f977-039f-4b1b-8c4c-ecee8ea3ff2f 1048
+++ _systemd_osc_context_escape /home/aruntej
+++ echo /home/aruntej
+++ sed -e 's/\\/\x5c/g' -e 's/;/\x3b/g' -e 's/[[:cntrl:]]/\0/g'
+++ printf '\033]3000;start=%,64s%;type=command;cwd=%,255s\033\'' 8a10e1e3-88d3-4e27-b619-74f1936b4372 'machineid=d003082ce67e438f87017759e4d73db3;user=aruntej;hostname=LAPTOP-DGRT8C5G;bootid=3387f977-039f-4b1b-8c4c-ecee8ea3ff2f;pid=00000000000000001048' /home/aruntej
+ set +x

```

```

aruntej@LAPTOP-DGRT8C5G:~$ echo "Special characters: $ @ # % & *"
Special characters: $ @ # % & *
aruntej@LAPTOP-DGRT8C5G:~$ name="Arun Tej"
aruntej@LAPTOP-DGRT8C5G:~$ echo "$name"
Arun Tej
aruntej@LAPTOP-DGRT8C5G:~$ printf '<%s>\n' "$name"
<Arun Tej>
aruntej@LAPTOP-DGRT8C5G:~$ empty=""
aruntej@LAPTOP-DGRT8C5G:~$ printf '<%s>\n' "$empty"
<>
aruntej@LAPTOP-DGRT8C5G:~$ mkdir "My Ubuntu Project"
aruntej@LAPTOP-DGRT8C5G:~$ cd "My Ubuntu Project"
aruntej@LAPTOP-DGRT8C5G:~/My Ubuntu Project$ cd ..
aruntej@LAPTOP-DGRT8C5G:~$ rmdir "My Ubuntu Project"
aruntej@LAPTOP-DGRT8C5G:~$ nano quotes.sh
aruntej@LAPTOP-DGRT8C5G:~$ chmod +x quotes.sh
aruntej@LAPTOP-DGRT8C5G:~$ ./quotes.sh
----- SINGLE QUOTES -----
$name
Welcome to Ubuntu
Today is $(date)
Hello          Ubuntu          Linux

----- DOUBLE QUOTES -----
My name is Aruntej
I am studying Computer Science
Current directory: /home/aruntej
Current date: Tue Sep 22 04:13:46 UTC 2026

With double quotes:
<Ubuntu Linux Lab>

Without double quotes:
<Ubuntu>
<Linux>
<Lab>
aruntej@LAPTOP-DGRT8C5G:~$ █

```